Week 6, Lecture 1
Jaynes–Cummings model

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Office hours: **This Week Thursday 1-2 pm**

Announcements

This is the last week of consultation for the quantum optics part of the course.

Office hours will be held from 1.00 to 2.00 pm this Thursday (note change from previous weeks)

Assessment reminder

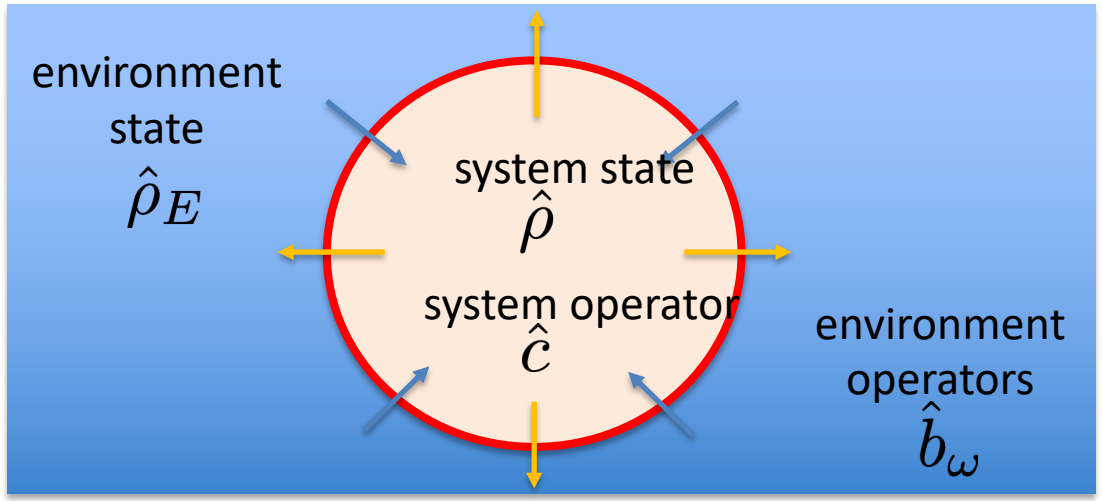
Reminder that assignment 2 is due 04/09/26 at 4pm

Common FAQs about assessment submission process:

- After submission: Click on the Download Digital Receipt icon to download a copy of the digital receipt as proof that you have submitted your assignment.
- Per the ECP: Feedback, grades and solutions will be posted two weeks after the due date to allow for extensions.
- If there are any issues with the submission process for this assignment, please email me.

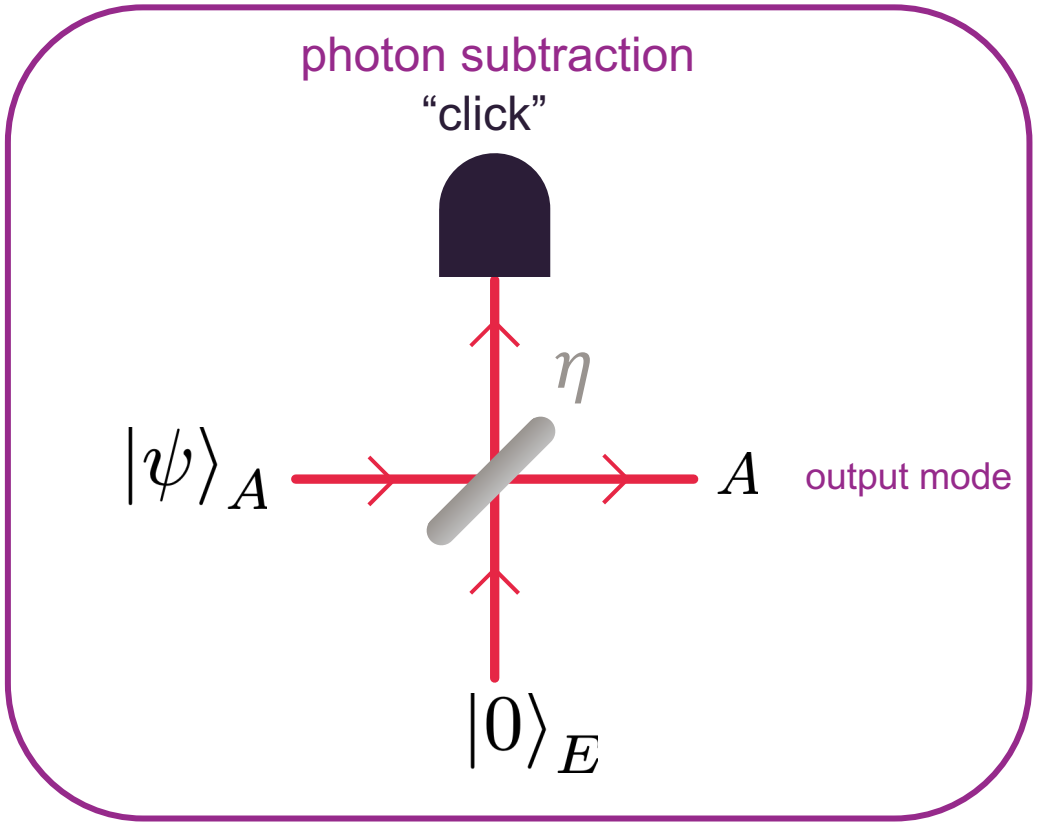
Review

What you learned last time…



What does the environment do?

How does this evolution develop in time?



The Jaynes–Cummings model

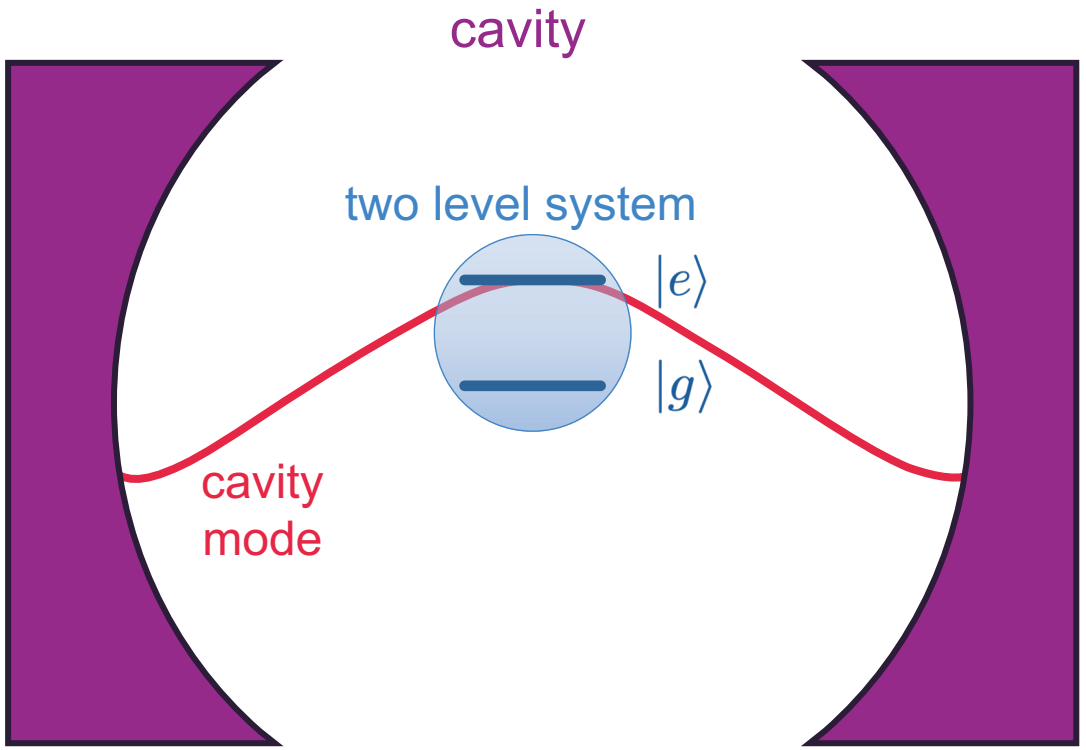
We have studied how to describe a **quantised electromagnetic field**.

Now we ask:

How does a quantised field interact with an atom?

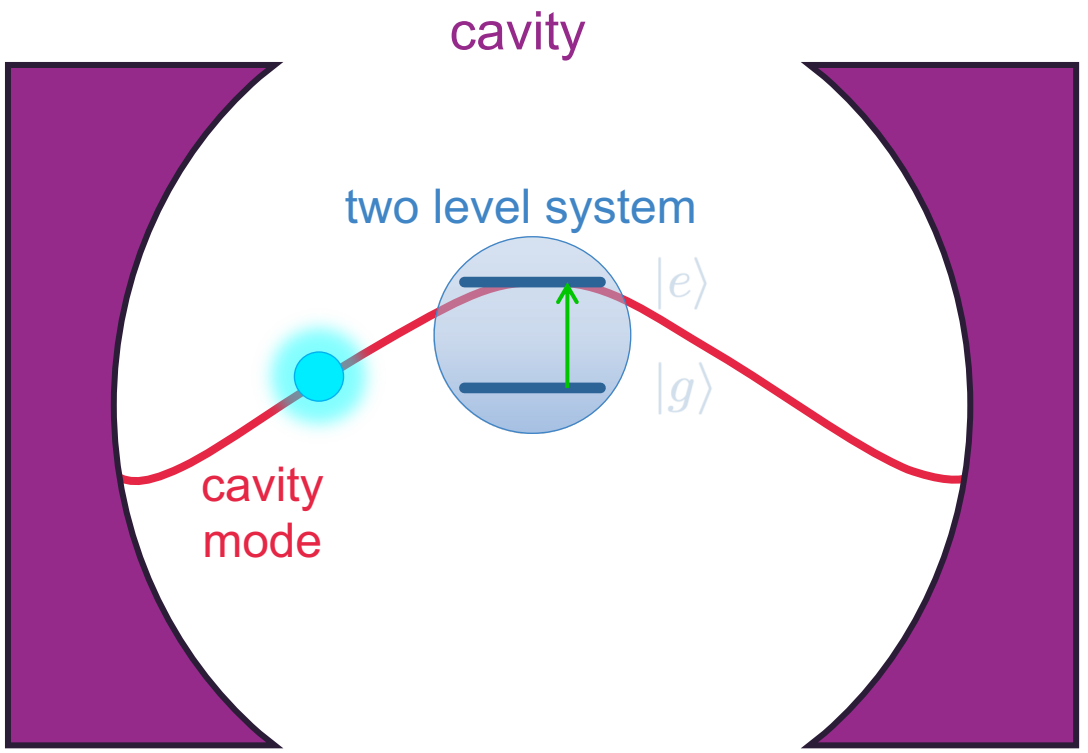
We will build the simplest useful model from two ingredients:

- an atom represented by two energy levels;
- a single quantised mode of the electromagnetic field.

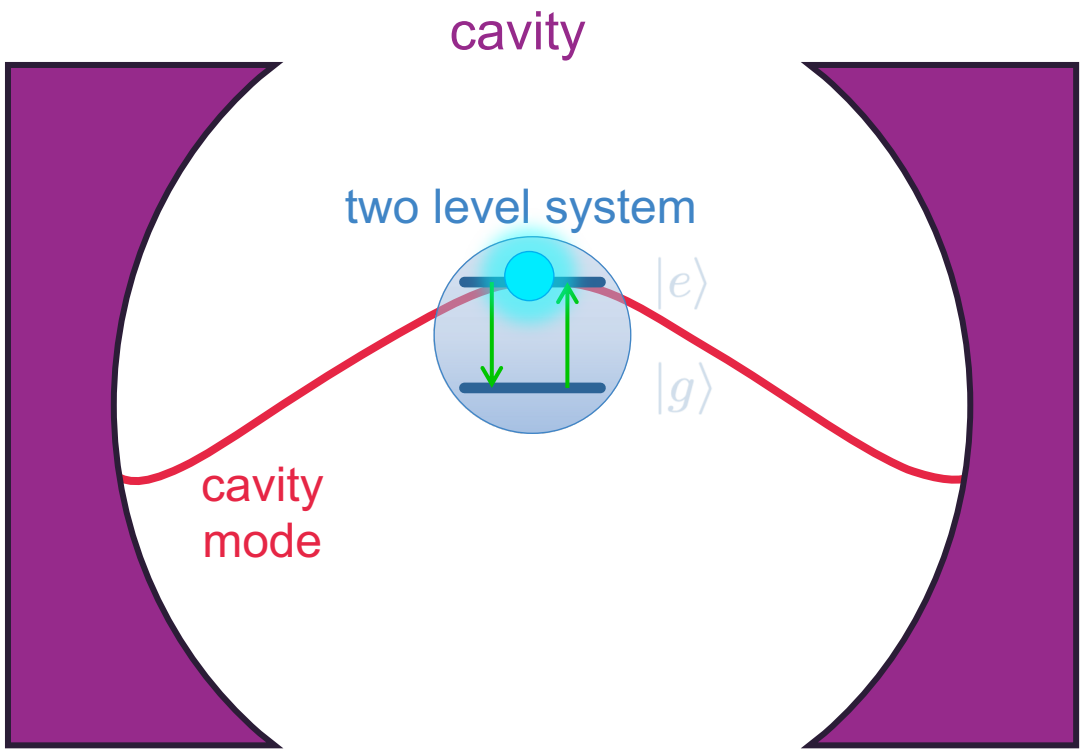


Together, these form the **Jaynes–Cummings model**.

The Jaynes–Cummings model



The Jaynes–Cummings model



What you will learn today

How the atom–field interaction Hamiltonian is constructed

Why particular atom–photon states become coupled

How an excitation oscillates coherently between matter and light

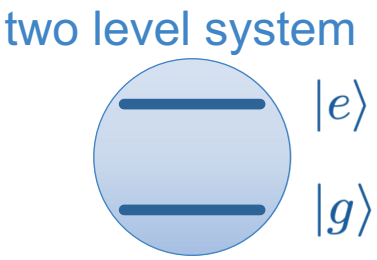
Modelling an atom as a two-level system

A real atom has many energy levels, but light close to one transition may interact predominantly with two of them:

$$|g\rangle \quad \text{ground state}, \qquad |e\rangle \quad \text{excited state}.$$

Their energy separation defines the atomic transition frequency:

$$E_e - E_g = \hbar\omega_a.$$



Choosing $E_g = 0$, the atomic Hamiltonian is

$$\boxed{H_{\mathrm{a}} = \hbar\omega_a |e\rangle\langle e|.}$$

The two-level approximation is appropriate when other atomic transitions are sufficiently far from resonance.

Atomic raising and lowering operators

Define the atomic raising and lowering operators:

$$\sigma_+ = |e\rangle\langle g|, \qquad \sigma_- = |g\rangle\langle e|.$$

Their actions are

$$\sigma_+|g\rangle = |e\rangle, \qquad \sigma_-|e\rangle = |g\rangle.$$

while

$$\sigma_+|e\rangle = 0, \qquad \sigma_-|g\rangle = 0.$$

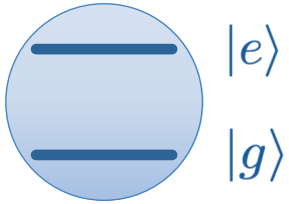
The atomic excitation operator is

$$\sigma_+\sigma_- = |e\rangle\langle e|.$$

so the atomic Hamiltonian can be written as

$$\boxed{H_{\mathrm{a}} = \hbar\omega_a\sigma_+\sigma_-.}$$

two level system

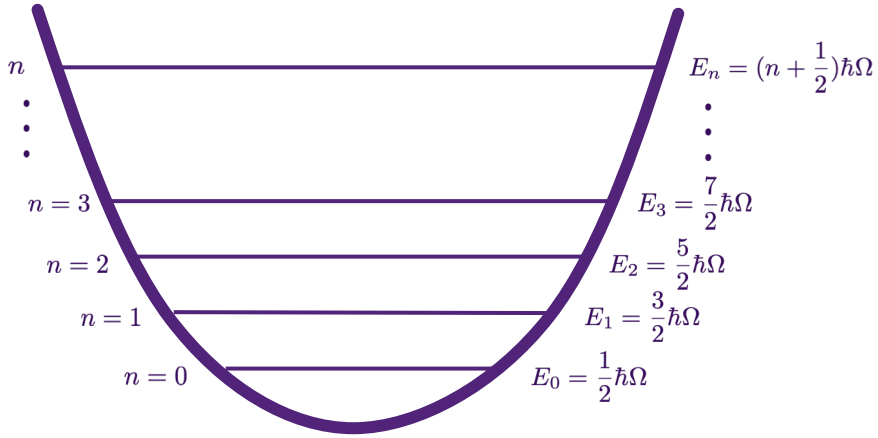


The other ingredient: a single field mode

Consider one quantised cavity mode with angular frequency ω_c.

Its annihilation and creation operators act as

$$a|n\rangle = \sqrt{n}\,|n-1\rangle, \qquad a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle.$$



Ignoring the constant vacuum energy, the field Hamiltonian is

$$\boxed{H_\mathrm{f} = \hbar\omega_c a^\dagger a.}$$

The photon-number operator is

$$\hat{n} = a^\dagger a, \qquad \hat{n}|n\rangle = n|n\rangle.$$

Joint atom–field states

The combined system is described using tensor-product states:

$$|g,n\rangle \equiv |g\rangle \otimes |n\rangle, \qquad |e,n\rangle \equiv |e\rangle \otimes |n\rangle.$$

Without an interaction, the total Hamiltonian is

$$H_0 = \hbar\omega_a \sigma_+\sigma_- + \hbar\omega_c a^\dagger a.$$

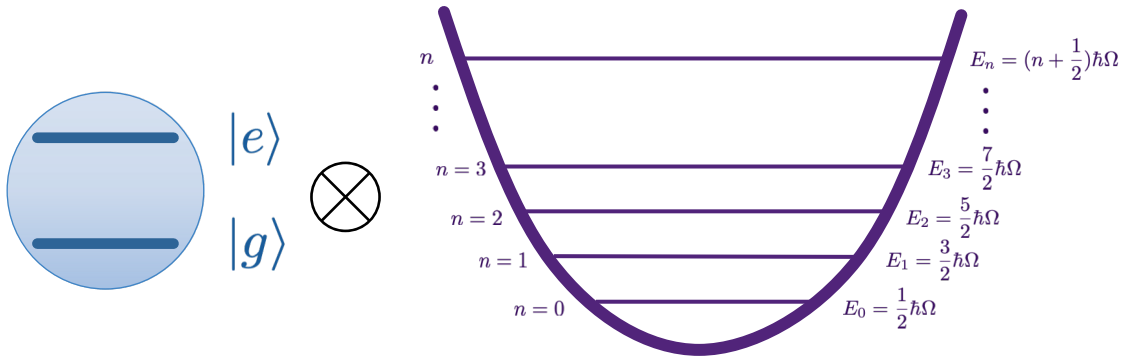
Therefore,

$$H_0|g,n\rangle = n\hbar\omega_c|g,n\rangle.$$
$$H_0|e,n\rangle = \hbar(\omega_a + n\omega_c)|e,n\rangle.$$

When $\omega_a \approx \omega_c$, the states

$$\boxed{|e,n\rangle \quad \text{and} \quad |g,n+1\rangle}$$

have approximately the same energy.



The electric-dipole interaction

In the electric-dipole approximation, the atom–field interaction is

$$H_{\text{int}} = -\hat{\boldsymbol{d}} \cdot \hat{\boldsymbol{E}}.$$

For a two-level atom,

$$\hat{\boldsymbol{d}} \propto \sigma_+ + \sigma_-.$$

and for a single field mode,

$$\hat{\boldsymbol{E}} \propto a + a^\dagger.$$

Absorbing the dipole matrix element and field amplitude into a coupling strength g,

$$H_{\text{int}} = \hbar g(\sigma_+ + \sigma_-)(a + a^\dagger).$$

Expanding gives four possible terms:

$$\boxed{H_{\text{int}} = \hbar g \left(a\sigma_+ + a^\dagger\sigma_- + a^\dagger\sigma_+ + a\sigma_-\right).}$$

The rotating-wave approximation

$$H_{\text{int}} = \hbar g \left(a\sigma_+ + a^\dagger \sigma_- + a^\dagger \sigma_+ + a\sigma_- \right).$$

Each interaction term describes a different process:

Term	Process
$a\sigma_+$	A photon is absorbed and the atom is excited
$a^\dagger\sigma_-$	The atom de-excites and creates a photon
$a^\dagger\sigma_+$	A photon and an atomic excitation are created together
$a\sigma_-$	A photon and an atomic excitation are removed together

In the interaction picture,

$$a\sigma_+ \propto e^{i(\omega_a - \omega_c)t}, \qquad a^\dagger\sigma_- \propto e^{-i(\omega_a - \omega_c)t}.$$

whereas

$$a^\dagger\sigma_+ \propto e^{i(\omega_a + \omega_c)t}, \qquad a\sigma_- \propto e^{-i(\omega_a + \omega_c)t}.$$

Near resonance, the latter terms oscillate rapidly and average to approximately zero:

$$H_{\text{int}}^{\text{RWA}} = \hbar g \left(a\sigma_+ + a^\dagger \sigma_- \right).$$

The Jaynes–Cummings Hamiltonian

Combining the free atom, free field and interaction terms gives

$$\boxed{H_{\mathrm{JC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a \sigma_+\sigma_- + \hbar g\left(a\sigma_+ + a^\dagger\sigma_-\right).}$$

The three contributions describe:

$$\underbrace{\hbar\omega_c a^\dagger a}_{\text{cavity field}} + \underbrace{\hbar\omega_a \sigma_+\sigma_-}_{\text{two-level atom}} + \underbrace{\hbar g\left(a\sigma_+ + a^\dagger\sigma_-\right)}_{\text{excitation exchange}}.$$

g: the single-photon coupling strength between the atom and cavity mode.

Define the atom–cavity detuning as

$$\boxed{\Delta = \omega_a - \omega_c.}$$

The atom and cavity are **on resonance** when

$$\Delta = 0 \qquad \Longleftrightarrow \qquad \omega_a = \omega_c.$$

The total excitation number is conserved

Define the total excitation-number operator

$$\boxed{\hat{N}_{\mathrm{exc}} = a^{\dagger}a + \sigma_{+}\sigma_{-}.}$$

It counts photons plus atomic excitations:

$$\hat{N}_{\mathrm{exc}}|e,n\rangle = (n+1)|e,n\rangle.$$

$$\hat{N}_{\mathrm{exc}}|g,n+1\rangle = (n+1)|g,n+1\rangle.$$

For the Jaynes–Cummings Hamiltonian,

$$\boxed{[H_{\mathrm{JC}}, \hat{N}_{\mathrm{exc}}] = 0.}$$

Therefore, the interaction can couple only states with the same total number of excitations:

$$|e,n\rangle \longleftrightarrow |g,n+1\rangle.$$

The state $|g,0\rangle$ is uncoupled.

Excitation manifolds

The Hilbert space separates into independent subspaces:

$$\mathcal{H}_0 = \mathrm{span}\{|g,0\rangle\}.$$

and, for $n = 0,1,2,\ldots$,

$$\boxed{\mathcal{H}_{n+1} = \mathrm{span}\left\{|e,n\rangle,\ |g,n+1\rangle\right\}.}$$

Within each subspace, the interaction acts as

$$a^\dagger \sigma_- |e,n\rangle = \sqrt{n+1}\,|g,n+1\rangle.$$
$$a\sigma_+ |g,n+1\rangle = \sqrt{n+1}\,|e,n\rangle.$$

The effective coupling is therefore

$$\boxed{g_n = g\sqrt{n+1}.}$$

Hamiltonian within an excitation manifold

Use the ordered basis

$$\{|e,n\rangle,\ |g,n+1\rangle\}\,.$$

Within this subspace, the Hamiltonian becomes

$$H_{n+1} = \hbar \begin{pmatrix} n\omega_c + \omega_a & g\sqrt{n+1} \\ g\sqrt{n+1} & (n+1)\omega_c \end{pmatrix}.$$

Using $\Delta = \omega_a - \omega_c$, this can be written as

where

$$\boxed{H_{n+1} = E_{\mathrm{av}} I + \frac{\hbar}{2} \begin{pmatrix} \Delta & 2g\sqrt{n+1} \\ 2g\sqrt{n+1} & -\Delta \end{pmatrix}.}$$

$$E_{\mathrm{av}} = \hbar \left[\left(n + \frac{1}{2} \right) \omega_c + \frac{\omega_a}{2} \right].$$

Finding the dressed states

On resonance, $\Delta = 0$, after removing the common energy $E_{\text{av}}I$, the Hamiltonian is

$$H'_{n+1} = \hbar g\sqrt{n+1}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Write an eigenstate as $|\psi\rangle = c_e|e,n\rangle + c_g|g,n+1\rangle \quad \longleftrightarrow \quad \begin{pmatrix} c_e \\ c_g \end{pmatrix}.$

The eigenvalue equation reduces to

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} c_e \\ c_g \end{pmatrix} = \mu\begin{pmatrix} c_e \\ c_g \end{pmatrix}.$$

which gives

$$c_g = \mu c_e, \qquad c_e = \mu c_g \quad \Longrightarrow \quad \mu^2 = 1.$$

Therefore

$$\mu = \pm 1, \qquad c_g = \pm c_e.$$

Finding the dressed states

$$\mu = \pm 1, \qquad c_g = \pm c_e.$$

After normalisation, $c_e = 1/\sqrt{2}$, giving the dressed states

$$\boxed{|n, \pm\rangle = \frac{|e, n\rangle \pm |g, n+1\rangle}{\sqrt{2}}.}$$

Their energies are

$$\boxed{E_{n,\pm} = \hbar(n+1)\omega_c \pm \hbar g\sqrt{n+1}.}$$

Hence, the dressed-state splitting is

$$E_{n,+} - E_{n,-} = 2\hbar g\sqrt{n+1}.$$

Why are they called dressed states?

Without the interaction, the atom and field have separate **bare states**:

$$|e, n\rangle, \qquad |g, n+1\rangle.$$

When the interaction is included, these bare states combine to form the eigenstates of the complete atom–field system:

$$|n, \pm\rangle = \frac{|e, n\rangle \pm |g, n+1\rangle}{\sqrt{2}}.$$

- The atom is said to be **dressed by the electromagnetic field**.
- The dressed states describe the atom and field together, rather than separately.
- The interaction changes both the states and their energies.

Time evolution from an excited atom

Suppose the atom begins excited with n photons in the cavity:

$$|\psi(0)\rangle = |e, n\rangle = \frac{|n, +\rangle + |n, -\rangle}{\sqrt{2}}.$$

Each dressed state evolves with its own phase:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(e^{-iE_{n,+}t/\hbar} |n, +\rangle + e^{-iE_{n,-}t/\hbar} |n, -\rangle \right).$$

Substituting the dressed states and their energies gives

$$|\psi(t)\rangle = e^{-i(n+1)\omega_c t} \left[\cos\left(g\sqrt{n+1}\, t\right) |e, n\rangle - i \sin\left(g\sqrt{n+1}\, t\right) |g, n+1\rangle \right].$$

Ignoring the physically irrelevant overall phase,

$$\boxed{|\psi(t)\rangle = \cos\left(g\sqrt{n+1}\, t\right) |e, n\rangle - i \sin\left(g\sqrt{n+1}\, t\right) |g, n+1\rangle.}$$

Jaynes–Cummings Rabi oscillations

The probability of finding the atom excited is

$$P_{e,n}(t) = |\langle e, n|\psi(t)\rangle|^2 = \cos^2\left(g\sqrt{n+1}\,t\right).$$

The probability of finding the atom in its ground state with one additional photon is

$$P_{g,n+1}(t) = |\langle g, n+1|\psi(t)\rangle|^2 = \sin^2\left(g\sqrt{n+1}\,t\right).$$

Writing these probabilities in the conventional form

$$P_{e,n}(t) = \cos^2\left(\frac{\Omega_n t}{2}\right), \qquad P_{g,n+1}(t) = \sin^2\left(\frac{\Omega_n t}{2}\right).$$

the Rabi angular frequency is

$$\boxed{\Omega_n = 2g\sqrt{n+1}.}$$

Complete transfer of the excitation occurs at

$$t_{\text{swap}} = \frac{\pi}{2g\sqrt{n+1}} = \frac{\pi}{\Omega_n}.$$

Rabi oscillations: coherent, periodic transfers of population between two coupled quantum states.

Vacuum Rabi oscillations

Suppose the cavity initially contains no photons and the atom is excited:

$$|\psi(0)\rangle = |e,0\rangle.$$

The interaction remains nonzero because

$$a^\dagger \sigma_- |e,0\rangle = |g,1\rangle.$$

The resulting evolution is

$$\boxed{|\psi(t)\rangle = \cos(gt)|e,0\rangle - i\sin(gt)|g,1\rangle.}$$

The corresponding probabilities are

$$P_{e,0}(t) = \cos^2(gt), \qquad P_{g,1}(t) = \sin^2(gt).$$

with vacuum Rabi angular frequency $\boxed{\Omega_0 = 2g.}$

- The atom emits one photon into the initially empty cavity.
- The photon can subsequently be reabsorbed by the atom.
- The energy comes from the initially excited atom—not from the vacuum.

Coherent exchange versus spontaneous emission

An ideal single-mode cavity produces reversible excitation exchange:

$$P_e(t) = \cos^2(gt).$$

For an atom emitting into many free-space modes, the excited-state population instead decays approximately as $P_e(t) = e^{-\Gamma t}$, where Γ is the spontaneous-emission rate.

Ideal cavity	**Free space**
One selected field mode	A continuum of field modes
The photon remains available	The photon propagates away
Coherent reabsorption is possible	Reabsorption is unlikely
Reversible oscillations	Approximately irreversible decay

In a real cavity, vacuum Rabi oscillations are visible when the coherent coupling is stronger than the relevant losses:

$$g \gg \kappa, \gamma,$$

where κ is the cavity loss rate and γ is the atomic decay rate.

Vacuum Rabi splitting

In the one-excitation manifold, the resonant dressed states are

$$|0,\pm\rangle = \frac{|e,0\rangle \pm |g,1\rangle}{\sqrt{2}}.$$

Their energies are

$$E_{0,\pm} = \hbar\omega_c \pm \hbar g.$$

A weak probe can therefore excite the system at two frequencies:

$$\omega_\pm = \omega_c \pm g.$$

The separation between these two resonances is

$$\boxed{\omega_+ - \omega_- = 2g.}$$

This appearance of two spectral peaks is called **vacuum Rabi splitting**.

Without coupling: one resonance at ω_c.

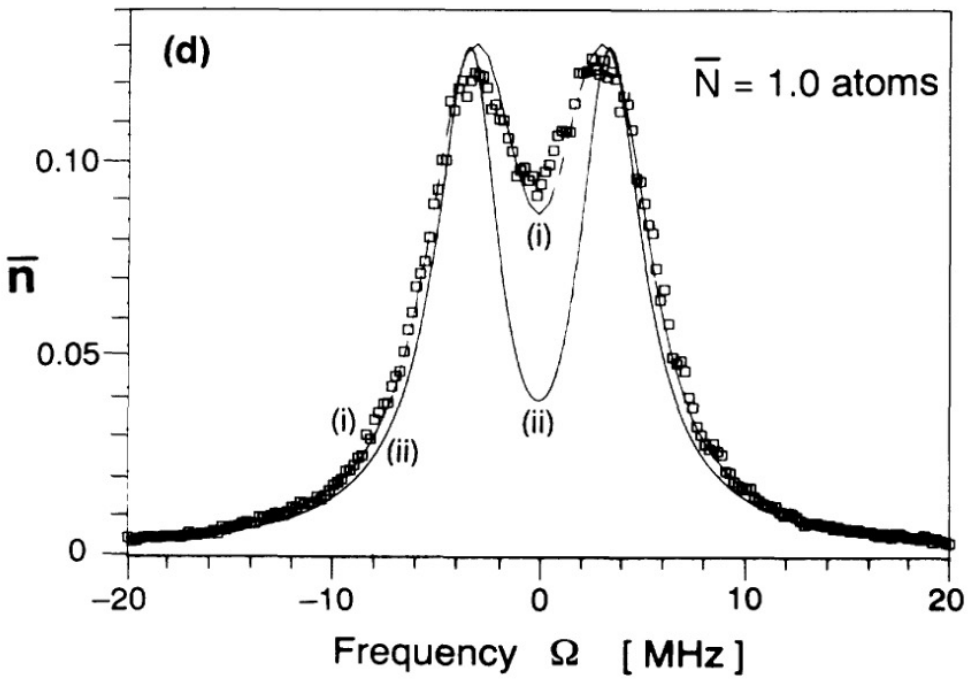
With strong coupling: two dressed-state resonances.

The splitting is resolvable when it exceeds the spectral linewidths.

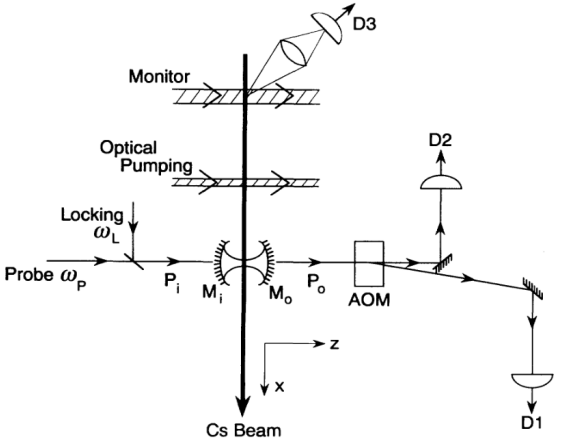
Vacuum Rabi splitting in the mean-one-atom regime

Experimental atom–cavity spectrum showing vacuum Rabi splitting for an average of one atom.

Cs atom in an optical cavity



R. J. Thompson, G. Rempe and H. J. Kimble, *Phys. Rev. Lett.* **68**, 1132 (1992).





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Observation of Normal-Mode Splitting for an Atom in an Optical Cavity

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(Received 4 November 1991)

An investigation of the spectral response of a small collection of two-state atoms strongly coupled to the field of a high-finesse optical resonator is described for mean number $\bar{N} \leq 10$ atoms. For weak excitation, a coupling-induced normal-mode splitting is observed even for one intracavity atom, representing a direct spectroscopic measurement of the so-called vacuum Rabi splitting for the atom-cavity system.

PACS numbers: 42.50.Fx, 32.80.−t, 42.65.Pc

Much of modern research in optical physics has as its cornerstone the interaction Hamiltonian describing the dipole coupling of a "two-state" atom to the electromagnetic field. In the preponderance of cases, as in the usual theories of the laser [1,2], of resonance fluorescence [3], and of optical bistability [4], the atom-field interaction can be described in a regime of weak coupling, for which the rate β for irreversible decay into the continuum of degrees of freedom of the electromagnetic field is much larger than the frequency scale g associated with reversible evolution in the interaction of any one mode of the field with the atom. Although weakly coupled systems can be strongly driven to produce coherent excitation with characteristic frequency $\chi \gg \beta$ (with χ proportional to the amplitude of excitation), the system remains weakly coupled in that the interaction frequency g for a single quantum is negligible. Note that χ becomes $g\sqrt{n}$ in the limit that the coupling frequency $g \to 0$ and the number of quanta $n \to \infty$.

In contrast to this usual circumstance of weak coupling, recently it has become possible in research in the area of cavity quantum electrodynamics [5] to achieve a situation of strong coupling for which $g \gg \beta$ and for which a photon emitted by an atom into a cavity mode is likely to be repeatedly absorbed and reemitted before irreversibly escaping into the environment [6–8]. This oscillatory exchange of excitation between atom and cavity mode arises from a ubiquitous normal-mode splitting in the eigenvalue spectrum of the interaction Hamiltonian for the strongly coupled system which was described long ago by Jaynes and Cummings [9] for a single intracavity atom ($\bar{N} = 1$) and by Tavis and Cummings [10] for $N > 1$. Although Jaynes-Cummings dynamics have been investigated for $N \approx 1$ and a few photons [6b], and while mode splittings have been studied previously for $N > 1$ [11–14] and quite recently for $N \approx 3$ [15], apparently the so-called vacuum Rabi splitting [16,17] of the first excited state for a single atom coupled to a cavity mode has not been similarly observed. With this backdrop, we report in this Letter an optical experiment where the weak-field normal-mode splitting for an atom-cavity system is investigated by direct spectroscopic measurements. In particular, for the case of average intracavity atomic number $\bar{N} \approx 1$, we record the response of the system to a variable-frequency input probe and observe a distinctive doublet symmetrically split by $\pm g$ about the otherwise common resonance frequency of atom and cavity. Because our measurements are carried out in an atomic-beam apparatus with fluctuations both in the number of atoms within the cavity-mode volume and in the atomic spatial positions, we present results from numerical simulations which extend the standard theory to approximate our situation. The agreement between this extended theory and the measured spectral responses is quite good and substantiates our observation of the normal-mode or vacuum Rabi splitting for a single intracavity atom.

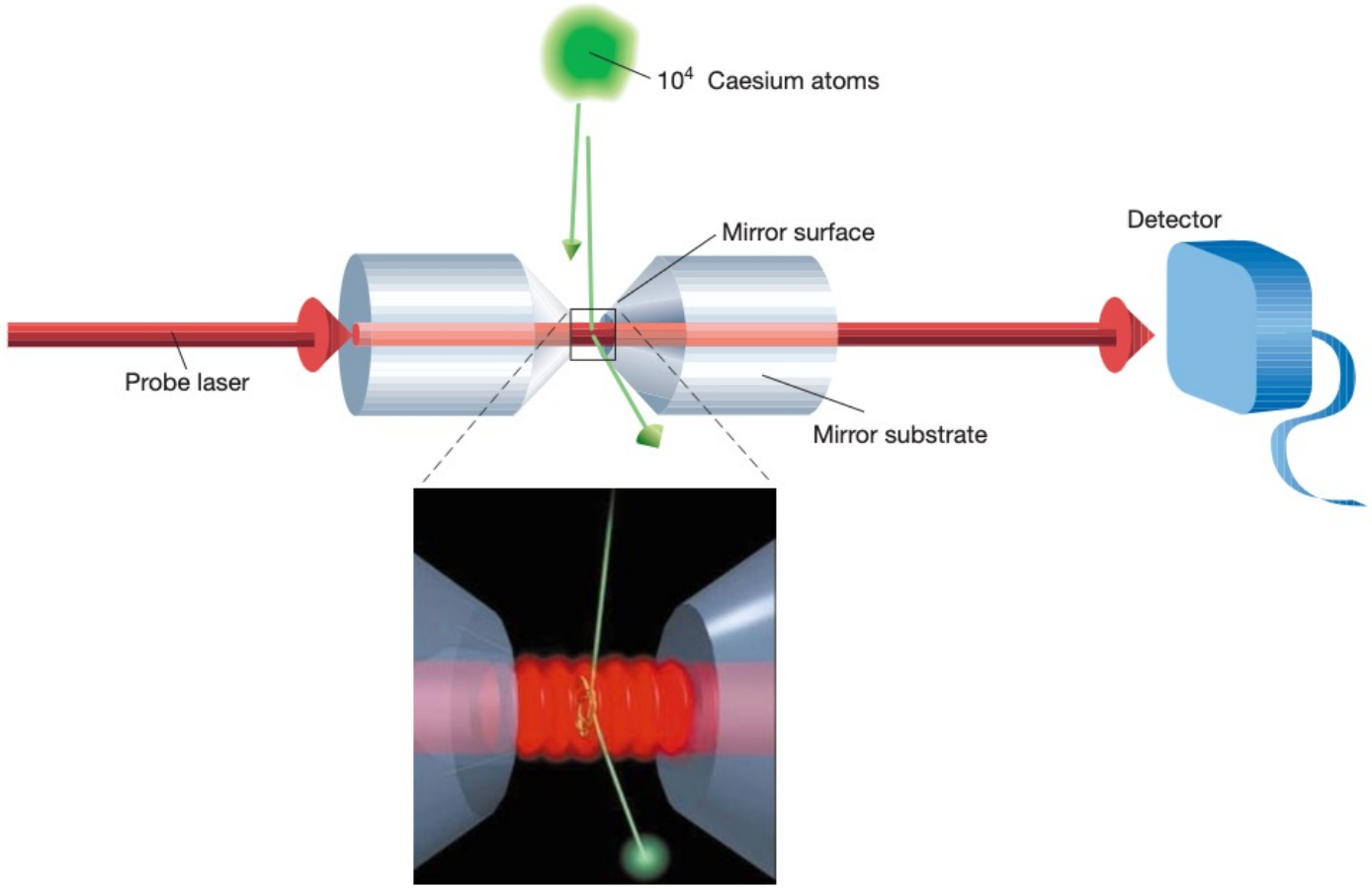
Our investigation of the structure of the atom-cavity system is restricted to a regime of weak excitation and consequently equivalent descriptions follow either from the quantum master equation or from the semiclassical Maxwell-Bloch equations, since in this domain the system behaves as coupled linear oscillators. Quantum mechanically the basis is limited to states with only zero or one quantum of excitation so that the coupling-induced splitting of the first excited state of the Jaynes-Cummings "molecule" is probed, while semiclassically the intracavity intensity is held much smaller than the saturation intensity so that the linear susceptibility of the composite entity is probed. In either case the response of the system is given in terms of the eigenvalues for the normal modes, which are of course independent of the particular excitation scheme and initial conditions. Our choice is to measure the steady-state transmission function $T(\omega_p)$ for a weak external probe field of variable frequency ω_p. $T(\omega_p)$ is found from the ratio of transmitted (P_t) to incident (P_i) power, and for the case of coincident cavity ω_c and atomic ω_a frequencies is given by [4,17]

$$T(\omega_p) = T_0 \left| \frac{\kappa[\gamma_\perp + i(\omega_0 - \omega_p)]}{(\omega_p - \lambda_+)(\omega_p - \lambda_-)} \right|^2, \quad (1)$$

where T_0 is the peak transmission of the empty cavity and κ is the cavity-field decay rate. Since the solid angle for our cavity mode is small ($\sim 10^{-4}$ sr), the rates $(\gamma_\perp, \gamma_\parallel)$ for the decay of the atomic polarization and inversion are as in free space, with $\gamma_\parallel = 2\gamma_\perp \approx \frac{1}{2}$ ns. The eigenvalues $\lambda_\pm$ describe the (collective) normal modes formed from the "cavity-field oscillator" and the "atomic-polarization oscillator" and are given by $\lambda_\pm = (\omega_a \pm \Omega_0) - (\gamma_\perp + \kappa)/2$, with $\omega_p = \omega_a \equiv \omega_0$ and the frequency $\Omega_0 \equiv [g^2 N - (\gamma_\perp - \kappa)^2/4]^{1/2}$ specifying a nor-

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Measuring the cavity transmission spectrum



Scan the probe frequency ω_p across the atom–cavity resonance.

At each frequency, measure the transmitted photon rate.

C. Monroe, “Quantum information processing with atoms and photons”, *Nature* **416**, 238–246 (2002), Fig. 7.

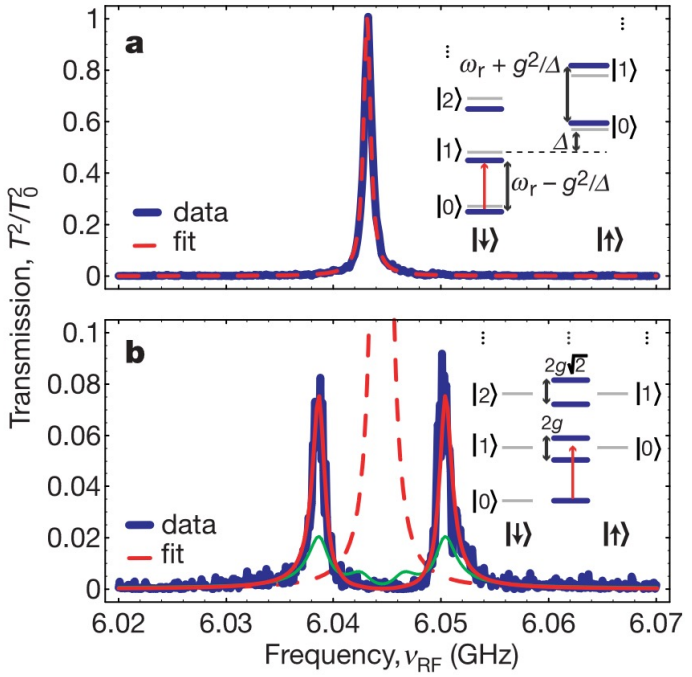
Jaynes–Cummings physics in a superconducting qubit

This experiment implements the same Jaynes–Cummings model in **circuit QED**

It uses a superconducting qubit, often described as an artificial atom, and an on-chip superconducting microwave resonator

Far off resonance, $|\Delta| \gg g$, the spectrum is dominated by one cavity-like peak.

On resonance, $\Delta = 0$, resolvable strong coupling produces two dressed-state peaks separated by approximately 2g



Wallraff, A., Schuster, D., Blais, A. *et al.* Strong coupling of a single photon to a superconducting qubit using circuit quantum electrodynamics. *Nature* **431**, 162–167 (2004).

Summary

One atom and one mode exchange excitations coherently

The Jaynes–Cummings Hamiltonian describes a two-level atom coupled to one quantised cavity mode:

$$H_{\mathrm{JC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a \sigma_+\sigma_- + \hbar g\left(a\sigma_+ + a^\dagger\sigma_-\right).$$

Conservation of total excitation number divides the system into coupled pairs:

$$|e,n\rangle \longleftrightarrow |g,n+1\rangle, \qquad \Omega_n = 2g\sqrt{n+1}.$$

The interaction creates dressed atom–field states, producing Rabi oscillations and a measurable energy splitting. For the vacuum field,

$$|e,0\rangle \longleftrightarrow |g,1\rangle, \qquad \omega_+ - \omega_- = 2g.$$